

Yesterday 6:24 PM

Suppose six points are randomly selected on the surface of a sphere and connected to form a hexagon. What is the probability that this hexagon is a trefoil knot?

Worked for 1m 36s >

Assuming six points are chosen independently and uniformly on the sphere, then connected in a fixed cyclic order by straight chords, the probability is approximately

$$P(\text{trefoil}) \approx 0.0068 \approx 0.68\%.$$

A Monte Carlo experiment found 68 trefoils in 10,000 random hexagons. Thus, roughly 1 in 150 such hexagons is a trefoil; this is a numerical estimate, not a known exact closed-form probability. [MathOverflow discussion](#) ↗

If the connections are geodesic arcs lying on the sphere's surface, the probability is instead 0, since such a curve is unknotted.

I would like you to derive an exact value for this.

Worked for 39m 34s >

Assume the same model as before: six i.i.d. uniform points on S^2 , connected in a fixed cyclic order by straight chords.

$$\Pr(\text{trefoil}) = \frac{1}{16\pi^2}$$

Numerically,

$$\frac{1}{16\pi^2} = 0.006332573977646 \dots,$$

so the probability is approximately

$$0.6332574\%.$$

The derivation is as follows.

For six points on a sphere, the convex hull is almost surely one of two combinatorial types:

Convex-hull type	Trefoil cyclic orders among the 60 unoriented cycles
Octahedral	1
Biaugmented tetrahedral	0

The finite cycle classification follows from Calvo's six-stick knot criterion, which distinguishes trefoils using signed triangle piercings [Calvo's classification](#).



Thus

$$\Pr(\text{trefoil}) = \frac{1}{60} \Pr(\text{octahedral hull}).$$

Fix one vertex as the north pole and stereographically project the other five points to the plane. Their density is

$$f(x, y) = \frac{1}{\pi(1 + x^2 + y^2)^2}.$$

Let X be the probability mass, under this density, of the triangle formed by three projected points. A fixed vertex has degree 3 precisely when one of the $\binom{5}{3} = 10$ possible triangles contains the other two, so

$$x_3 = 10\mathbb{E}[X^2].$$

The irregular hull has exactly two degree-3 vertices, while the octahedral hull has none. Therefore

$$\Pr(\text{octahedral hull}) = 1 - 30\mathbb{E}[X^2].$$

To evaluate $\mathbb{E}[X^2]$, use the one-form

$$\omega = \frac{x \, dy - y \, dx}{1 + x^2 + y^2}, \quad d\omega = \frac{2 \, dx \, dy}{(1 + x^2 + y^2)^2}.$$

For a triangle with vertices a, b, c ,

$$X^2 = \frac{1}{4\pi^2} (I(a, b) + I(b, c) + I(c, a))^2,$$

where

$$I(a, b) = \int_a^b \omega.$$

Reflection symmetry makes the cross terms vanish. Writing the line through a, b as

$$a = pn + ut, \quad b = pn + vt, \quad v > u,$$

and then substituting $u = \sqrt{1+p^2} \tan \alpha, v = \sqrt{1+p^2} \tan \beta$, gives

$$\mathbb{E}[X^2] = \frac{3}{4\pi^2} \frac{2}{\pi} \left(\frac{4}{15} \right) \left(\frac{\pi^3}{12} - \frac{5\pi}{16} \right).$$

Hence

$$\mathbb{E}[X^2] = \frac{1}{30} - \frac{1}{8\pi^2}.$$

Therefore

$$\Pr(\text{octahedral hull}) = 1 - 30 \left(\frac{1}{30} - \frac{1}{8\pi^2} \right) = \frac{15}{4\pi^2},$$

and finally

$$\Pr(\text{trefoil}) = \frac{1}{60} \frac{15}{4\pi^2} = \boxed{\frac{1}{16\pi^2}}.$$

If you meant great-circle arcs lying on the sphere rather than straight chords, the probability is 0: a simple curve on S^2 is unknotted.